Yifei (Bruce) Li

Robot Learning | Safe Planning | VLA | World Model | Agentic Robotics | **Seeking Fulltime & PhD Opportunities**
+1 314 349 6762 | liyf@engineering.upenn.edu | liyifei1218.github.io | linkedin.com/in/liyifei1218

Education

Also applying for PhD stating Fall 2027
Prospective PhD Student in Robotics / ECE / Computer Science — *2027–2032 (planned)*

University of Pennsylvania - GRASP Laboratory — *Philadelphia, PA*
Master of Science in Robotics — *Aug 2025 - (Expected) May 2027*

- **GPA: 3.82/4.00**; **Relevant Coursework:** *Learning in Robotics, Projective Geometry and Machine Perception, Machine Learning, Kinematics, Dynamics, Control and Optimization, World Models, Deep Generative Models*

Washington University in St. Louis — *St. Louis, MO*
B.S. in Computer Engineering; B.S. in Computer Science; minor in Psychological & Brain Science — *Sept 2021 - May 2025*

- **Major GPA: 3.98/4.00**; **Overall GPA: 3.89/4.00**
- **Graduated Cum Laude**; 2024 Summer Undergraduate Research Award; Dean's List: Spring 2022 - Spring 2024

Research Experiences & Publications

World-Model Feedback for Safe Multimodal Robot Action Generation (Ongoing) — *Philadelphia, PA*
Graduate Researcher, Figueroa Robotics Lab, advised by **Dr. Nadia Figueroa** — *Sept 2026 - Present*

- Proposed world-model feedback to evaluate and refine diffusion-policy actions for mobile manipulation, targeting collision avoidance and continued task progress through coordinated base and arm motion.

Safety for Decentralized Multi-Robot Teams Using LLM Planners (Ongoing) — *Philadelphia, PA*
Graduate Researcher, advised by **Dr. George J. Pappas** — *Aug 2026 - Present*

- Built a benchmark for decentralized LLM-controlled robot teams in Isaac Sim. Showed that compromising one robot reduced three-robot coverage from 94.4% to 68.1–76.4% across direct, peer-message, and goal-hijack attacks.

Online RL for Flow-Based Robot Policies and VLA Distillation (Ongoing) — *Philadelphia, PA*
Graduate Researcher, advised by **Dr. Osbert Bastani** — *Jul 2026 – Present*

- Built an online flow-policy RL benchmark in PyTorch/Isaac Lab; reimplemented FlowDPG and SAC-Flow.
- Trained flow-based manipulation policy teacher from scratch to generate demonstrations for VLA fine-tuning and sim-to-real transfer

Memory-Aware Multi-Sensor Navigation in Dynamic Environments — *Philadelphia, PA*
Graduate Researcher, Figueroa Robotics Lab, advised by **Dr. Nadia Figueroa** — *Mar 2026 - Sep 2026*

- Proposed a memory-aware navigation stack integrating LIO state estimation, human tracking, persistent mapping, Gaussian-process distance fields, and adaptive MCBF-QP control for map-free navigation in concave, dynamic environments.
- Achieved 100% success across simulation and deployed end-to-end stack on ground platform with onboard LiDAR and Jetson Orin AGX
- **Presented at IROS 2026 workshop, best contribution award finalist; extended conference manuscript submitted to ICRA 2027**

Safe Planning in Unknown Environments using Conformalized Semantic Maps — *St. Louis, MO*
Undergraduate Researcher, Kantaros Lab, advised by **Dr. Yiannis Kantaros** — *Jan 2023 - Sep 2025*

- The proposed semantic planner for reach-avoid tasks integrates conformal prediction to quantify semantic map uncertainty without assuming noise distributions, achieving tasks with user-defined probability despite uncertain perception across extensive experiments.
- **Manuscript accepted to IEEE Robotics and Automation Letters (RA-L); presented at IROS 2026**

Projects

Reinforcement Learning for Autonomous High-Speed Quadcopter Racing — *Philadelphia, PA*
University of Pennsylvania — *Jan 2026 - May 2026*

- Trained PPO policy in Isaac Sim using rsl_rl for an autonomous quadcopter racing at competitive speed with sim2real transfer
- designed sparse-dominated reward, domain randomization, and reset strategy, sim2real with 90% performance and 98% success rate

Autonomous Pick-and-Place using 7-DOF Robot Arm — *Philadelphia, PA*
University of Pennsylvania — *Oct 2025 - Dec 2025*

- Developed a full-stack motion planning pipeline using custom FK and IK solvers with null space optimization for the Franka Emika Panda
- Use end-effector mounted camera to predict and manipulate dynamic objects, achieving 95% success in simulation and 70% on hardware

Supervised Learning with Reduced-Order HJB Seeding for Quadrotor Control — *Philadelphia, PA*
University of Pennsylvania — *Oct 2025 - Dec 2025*

- Implemented a HJB-seeded learning pipeline by solving discretized HJB on reduced 6D model
- Trained supervised policy from the HJB and deployed on a 12D quadrotor simulator for single object avoidance

Web Application to Help SMEs Pursue Digital Transformation — *Remote / Shanghai, China*
Intern - Backend Developer — *Jul 2023 - Sept 2023*

- Completed entire development process, incl. backend using Node.js, Java Spring Boot and SQL, and launching with 5+ initial customers

Skills

Robotics/ML Isaac Sim/Lab, VLAs, Generative Policies, Gazebo, PyTorch, FK/IK, ROS 1/2, SLAM, Sensor Fusion, Mujoco, SolidWorks, 3D Printing, CNC
Other Python, Java, C/C++, MATLAB, C#, HTML, Swift, JavaScript, SQL, LaTeX, Unity Engine, AWS, GCP